

RESEARCH INTERESTS

- Robot Foundation Models and Embodied AI
- Efficient AI Systems and Multimodal Model Serving
- Memory and Computation Reuse for Interactive AI

EDUCATION

2025–Present	Sungkyunkwan University <i>Integrated M.S./Ph.D. Program in Software</i> Department of Software Lab: CSI Lab Advisor: Honguk Woo GPA: 4.14 / 4.5	Suwon, Republic of Korea
2019–2024	Sungkyunkwan University <i>B.S. in Software</i> Department of Software GPA: 3.41 / 4.5	Suwon, Republic of Korea

RESEARCH EXPERIENCE

2024–Present	CSI Lab, Sungkyunkwan University <i>Graduate Researcher</i> • Conduct research on robot foundation models, embodied AI, and efficient AI systems. • Study efficient inference for Vision-Language-Action (VLA) and multimodal models, focusing on temporal redundancy and computation reuse. • Develop memory and cache mechanisms for long-running embodied agents, including structured KV reuse and function-level code-policy synthesis. • Investigate techniques to reduce redundant prefill and inference computation across sequential observations while keeping reused state consistent. • Research publications include HeMeR (NeurIPS 2026, Oral) and two ICML 2026 papers (Poster).	Suwon, Republic of Korea
2023–2024	VLDB Lab, Sungkyunkwan University <i>Undergraduate Research Assistant</i> • Conducted research on memory-management optimization for HTAP workloads. • Analyzed buffer-pool pollution caused by undo pages and explored system-level mechanisms to mitigate it. • Designed and evaluated MySQL/InnoDB, C++-based mechanisms for improving memory behavior under mixed transactional and analytical workloads. • Resulted in domestic conference publications at KDBC 2023 and KCC 2024.	Suwon, Republic of Korea

SELECTED ONGOING RESEARCH

Efficient Inference for Robot Foundation Models Study temporal redundancy, observation reuse, asynchronous inference, and stale-state mitigation for VLA models operating in continuous perception–action loops. The aim is to improve responsiveness while preserving consistency with changing observations.	Ongoing
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International Conferences (3)

- [1] **HeMeR: Heterogeneous Memory Reconciliation for Embodied Agents via Structured KV Reuse.**
Saehun Chun, Wonje Choi, Jinwoo Jang, TaeYoon Kwack, and Honguk Woo*.
Conference on Neural Information Processing Systems (NeurIPS), 2026. **Oral**.
- [2] **Functional Cache Grafting: Robust and Rapid Code-Policy Synthesis for Embodied Agents.**
Saehun Chun, Wonje Choi, Sera Choi, Sanghyun Ahn, and Honguk Woo*.
International Conference on Machine Learning (ICML), 2026. **Poster**.
- [3] **Efficient Skill Grounding via Code Refactoring with Small Language Models.**
Sera Choi, Wonje Choi, **Saehun Chun**, Daehee Lee, Chaeun Lee, Jooyoung Kim, and Honguk Woo*.
International Conference on Machine Learning (ICML), 2026. **Poster**.

Domestic Conferences (2)

- [1] **Undo Read Buffer: Preventing Buffer Pollution due to Undo Blocks in HTAP Workloads.**
Saehun Chun, Jonghyeok Park, Sang-Won Lee*, and Young Ik Eom.
Korea Computer Congress (KCC), 2024.
- [2] **Analysis of Buffer Pollution Caused by Undo Pages in HTAP Workloads.**
Saehun Chun, Bo-Hyun Lee, Kyong-Shik Lee, Sang-Won Lee*, and Young Ik Eom.
Korean DataBase Conference (KDBC), 2023.

RESEARCH PROJECTS

2024–Present

HiPER: High-Performance Exabyte Storage Systems*Project Manager / Graduate Researcher*

Samsung Electronics

- Develop efficient system support for large-scale foundation-model and embodied-agent workloads.
- Coordinate an industry-academic research collaboration with Samsung Electronics, including project execution, experiment planning, progress reporting, and technical deliverables.
- Conduct research on memory- and storage-efficient system support for large-scale AI workloads, focusing on cache reuse, data movement reduction, and performance analysis across heterogeneous memory/storage hierarchies.
- Develop and evaluate system prototypes and benchmarking methodologies for LLM/agent-serving workloads, analyzing how storage and memory-system design affects throughput, latency, scalability, and resource efficiency.

RESEARCH PROJECTS (CONTINUED)

Jan. 2024	Global Capstone Design <i>Project Manager</i> Sungkyunkwan University & Victoria University of Wellington • Applied machine learning to adaptively manage the MySQL buffer pool. • Designed and prototyped AI-driven buffer management for better memory efficiency. • Presented at Victoria University of Wellington and collaborated with local students.
2020	Industry-Academic Collaboration Project <i>Project Manager</i> SK M&N Service • Led a project with SK M&N Service to develop an STT-based video summarization system. • Conducted video/audio data preprocessing and trained machine learning models, including unsupervised approaches, for concise summary generation. • Technologies: Python, TensorFlow, Google Colab. Repository: github.com/thousae/SKmnService.

AWARDS AND HONORS

2025–2026	Graduate Research Fellowship , National Research Foundation of Korea.
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TEACHING EXPERIENCE

2026–Present	GenAI PowerUser Program Level 4 (Multi Agent framework), Samsung Electronics University <i>Hands-on Training Instructor</i>
Fall 2026	Artificial Intelligence Project, Sungkyunkwan University <i>Hands-on Training Instructor</i>
Spring 2026	System Programming, Sungkyunkwan University <i>Teaching Assistant</i>
Spring 2025	Mobile Application Programming, Sungkyunkwan University <i>Teaching Assistant</i>

SKILLS

Programming	Python, C/C++
ML / Systems	PyTorch, CUDA, Docker, Linux, Git
Research Tools	LaTeX, Weights & Biases, Hugging Face
Languages	Korean (Native), English (Fluent), Chinese (Basic)